

Basic Code Analysis: Async/Await & Thread Blocking Analysis

Duration : 30 Min.

Scenario

A Node.js event loop freezes because synchronous `fs.readFileSync` is used inside an HTTP handler. Create 'app.js' in `student_workspace/` using asynchronous `fs.promises.readFile`.

Target File Location & Creation

File to Create/Update: `student_workspace/app.js`

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Create 'app.js' inside 'student_workspace/'. Implement 'handleRequest' using `async/await` with '`fs.promises.readFile`'.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect the scenario input files in `student_workspace/`.
3. Create/update the file `student_workspace/app.js` and populate it with valid parameters and configurations.
4. Save your changes (`Ctrl + S` or `Cmd + S`).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your work by running `python run.py` in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	app.js exists in student_workspace/ with valid JS code	3 Marks
TC2	readFileSync removed from handler	4 Marks

TC3	async/await or promises readFile implemented	3 Marks
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Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.